Resampling Cannot See the User: Detecting User-Specific Hallucination in Interactive Agents with Probabilistic Commitments

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Abstract

An agent serving an individual must act on facts about that person it was never told: whether a scan is safe today, whether a sedative is needed. When it asserts such facts confidently and wrongly, nothing external can check it, because the truth lives with the user. We show that sampling-consistency detectors cannot tell users apart: conditional on the conversation, the model's resamples carry no information about which user it is wrong about, and the more consistently an agent fabricates, the closer their AUROC is to $1/2$. We propose Probabilistic Game Environments (PGE), a protocol in which the agent states each assumption as an explicit commitment with a confidence and the user's implicit reactions are recorded, and a label-event posterior that turns commitments and reactions into per-commitment and per-session risk. In a simulator where the truth is known, exact Bayes predictors split what is detectable into population risk, the chance that these commitments are wrong for a user with the same intake, and individual evidence from the user's reactions. In a pre-registered evaluation on fresh sessions with five LLM agents from four families, sampling detectors score **??** AUROC on user-specific hallucination, detectors on PGE observables reach up to [PENDING: max], and reactions add **??** AUROC beyond the population risk of the same commitments. With a simulated patient that answers in natural language, and with the agent re-reasoning after each answer, choosing what to ask by posterior risk [PENDING: clarification headline]. We report every pre-registered outcome, including those against our method.

Keywords: LLM agents; hallucination detection; uncertainty; clarification; implicit feedback.

1 Introduction

An agent that schedules imaging appointments must decide whether a patient may have a contrast scan today. The intake form says the patient had a recent treatment, but not when. A careful agent would say it does not know; many instead answer "safe, $p = 0.85$" and proceed. We call this a *user-specific hallucination*: a confident assertion about the current user whose truth depends on something the user never disclosed. It differs from the factual hallucinations that dominate the literature [10]: there is no document or search engine to check it against.

The leading black-box detectors resample the model and flag answers it does not reproduce [8, 16, 20]. That presumes hallucinations are stochastic. Fabricated user facts need not be: an underdetermined question maps the same context to the same guess every time. More fundamentally, two users who share a conversation (the same intake answers and, where the agent is shown them, the same reactions) are indistinguishable to the model, so nothing computed from the model and its context can tell which of them it is wrong about (Theorem 1). Context can still reveal *population* risk, for instance that "no sedation needed" is wrong for a third of patients, and we show that a well-calibrated context-only predictor can exploit it. Sampling-consistency scores of a self-consistent agent do not even do that (Corollary 1).

The way out is to look at the user rather than the model. In PGE the agent must externalize each assumption as a *commitment* $(\text{attribute}, \text{value}, p)$, and the user's reaction (accepting it, contradicting it, or ignoring it) is recorded. Reactions are noisy and partial, but they depend on the truth for *this* user, which no resample does: under a simple model the posterior of the hallucination event uses them exactly (Proposition 1), and less noisy reactions cannot hurt (Theorem 2). Our contributions:

1. **Why sampling fails, measured.** A within-context impossibility result and a dispersion bound, checked per commitment on five agents: on commitments whose truth depends on undisclosed user facts, resample disagreement predicts error at [PENDING: range over agents] AUROC, a single noisy reaction at [PENDING: range].

2. **A decomposition of detectability.** Exact Bayes predictors in a simulator with known truth separate the population risk of the commitments from the individual evidence in the reactions; the posterior given commitments and reactions bounds what any detector, with or without resamples, can achieve.

3. **A protocol and a posterior.** PGE's commitments and reactions make user-specific hallucination detectable; its label-event posterior gives per-commitment risk that is exact under a conditional-independence model (estimated by plug-in), which we use to choose what

to ask the user, measured with the agent re-reasoning after each answer.

4. **A pre-registered evaluation.** After an independent audit of our development results, we fixed the analysis, fixed a protocol bug, and generated fresh sessions with five agents from four families and a natural-language simulated patient. We report every registered hypothesis, including those that go against the posterior.

2 Related Work

Hallucination detection. Sampling-based detectors measure agreement across resamples [8, 16, 20]; semantic entropy is explicitly scoped to stochastic confabulations, not consistent errors [8]. Learned confidence models from a model's generations [31], white-box probes [2, 4, 15], and verification by the model's own questions [7] or by search [32] are all functions of the model given its context, so Theorem 1 applies to them. Consistent errors are documented in snowballing [35] and self-contradiction studies [21], and calibrated models must hallucinate on facts absent from training [12]; our setting is the per-user analogue.

Hallucination in agents. MIRAGE-Bench [36] taxonomizes agent actions unfaithful to instructions, history or observations; CAR-bench [14] tests whether agents admit missing information with a simulated user; AgentHallu [18] localizes the step that introduced an error. We study detection when the missing information is the user's own.

Clarification. When to ask has been framed through expected value of perfect information [25, 30], ambiguity detection [34], conformal prediction [26], and information-gain rewards on τ-bench [6, 33]. These reason about the agent's uncertainty before it acts; we use post-hoc commitment risk to choose which assertions to confirm, and measure the effect with the agent re-reasoning.

Implicit and noisy feedback. Implicit signals such as clicks are informative but biased [11]. Our estimation step inverts a noise channel, as in learning from noisy labels [1, 22, 28] and annotator models [5]. Agents that revise under user pushback raise sycophancy concerns [29]; our simulated user follows a long line of user simulation for dialogue [27].

3 Problem, Protocol, and Labels

Definition 1 (User-specific hallucination)**.** *A* commitment *asserts a value v for an attribute a of the current user with confidence* $p \in [0,1]$*. It is* user-dependent *if its true value is not determined by what the user has disclosed, and it is a user-specific hallucination if it is user-dependent, confident (*$p \geq \tau$*), and wrong.*

Protocol. A PGE session has four stages. (i) *Intake*: the user answers a few coarse questions; the agent sees only these answers. (ii) *Commitment*: at each turn the agent emits one commitment in a fixed format, `[attribute]: value, p=confidence`. (iii) *Reaction*: the user's implicit reaction $F_t \in \{-1,0,+1\}$ (contradicts, no reaction, accepts) is recorded and shown to the agent for its next six turns. (iv) *Detection and intervention*: a detector scores the session and its commitments; high-risk commitments trigger a confirmation question.

Labels. With confident set $H = \{t : p_t \geq \tau\}$, $\tau = 0.6$, the *composite* label L (Eq. (1)) marks a session in which at least half of the confident commitments are wrong, and is 0 if there are none. It mixes two kinds of error. Some attributes are determined by the intake through rules the agent was never shown (e.g. contrast is needed iff urgency is medium or high); errors there are guesses about the clinic, not about the user. The *user-specific* label L_U applies the same rule to the confident commitments that are user-dependent in that session, and is 0 when there are none. Whether a commitment is user-dependent is decided per session by enumerating the hidden profile's support: `scan_safe_now`, for example, depends on days since treatment only when a treatment was recent. L_U is our primary label; L is reported as secondary.

4 What Resampling and Feedback Can Reveal

Setup. A user u is drawn from a population. At turn t the agent conditions on a context X_t: the coarse intake $\phi(u)$, the question, and the interaction history, which in PGE displays earlier reactions; X denotes $(X_1, \ldots, X_T)$. It emits a commitment $c_t = (a_t, v_t, p_t)$ (attribute, value, reported confidence) from its policy $\pi(\cdot \mid X_t)$. Apart from the truth and the reactions, nothing depends on u except through X. The truth $y_a(u)$ is private to the user; the commitment is wrong if $w_t = \mathbf{1}[v_t \neq y_{a_t}(u)]$. With confident set $H = \{t : p_t \geq \tau\}$ and error fraction ρ, a session hallucinates iff

$$L = \mathbf{1}\Big[\, |H| \geq 1 \text{ and } \textstyle\sum_{t \in H} w_t \;\geq\; \lceil \rho\, |H| \rceil \,\Big]. \qquad (1)$$

The results below hold for any binary label determined by (C,u): a single w_t, or L_A, Eq. (1) with H replaced by $H_A = \{t \in H : a_t \in A\}$ for an attribute set A determined by X (for the user-specific label L_U, the attributes the intake does not determine). We write L.

A *context-and-sample detector* is any $S = h(X, C, Z, \xi)$ with ξ independent randomness and Z drawn, given (u, X, C), from a kernel $K(\cdot \mid X, C)$ that does not involve u: resamples at any temperature, token log-probabilities, hidden states. A *feedback detector* also sees reactions $F_t \in \{-1,0,+1\}$ (contradicts, none, accepts). By the Neyman–Pearson (NP) lemma [23], among all, possibly randomized, detectors that are functions of an observation O, thresholding $P(L = 1 \mid O)$ (randomized at ties) has the

highest true-positive rate at every false-positive rate, hence the highest AUROC; ties count $\frac{1}{2}$. Theorem 1 is the standard data-processing consequence of NP [3, 23], stated for this setting.

Theorem 1 (Self-consistency bound)**.** *Let $\eta_\varnothing = P(L = 1 \mid X, C)$ and S be a context-and-sample detector. Given (X, C), S is independent of u and L, so within a context it ranks hallucinating above clean sessions with probability exactly $\frac{1}{2}$. Across contexts, its ROC curve lies pointwise below that of the Bayes predictor $\eta_\varnothing$.*

Proof. Given (u, X, C), S depends only on (Z, ξ), whose law does not involve u, and L is a function of (C, u). So the law of S given (X, C) does not depend on L, $P(L = 1 \mid X, C, S) = \eta_\varnothing$, and NP applies to $O = (X, C, S)$. □

Since X contains the reactions the agent was shown, $\eta_\varnothing$ is not feedback-free, and detectors built from (X, C), such as context-only classifiers, can reflect them and exploit base rates across contexts. Resampling cannot add to (X, C) or separate users who share a context; a feedback detector reads the reactions directly, including the last. For dispersion scores, more is true:

Corollary 1 (Dispersion under systematic fabrication)**.** *Let S be a score of the commitments and their k resamples (agreement rate, semantic entropy, SelfCheck-style inconsistency) that takes a fixed value $s^\star$ whenever every resample equals its commitment, and $e_\ell = P(S \neq s^\star \mid L = \ell)$. Then*

$$\left|\mathrm{AUROC}(S) - \tfrac{1}{2}\right| \leq \tfrac{1}{2}\big(1 - (1 - e_0)(1 - e_1)\big) \leq \tfrac{1}{2}(e_0 + e_1).$$

If $s^\star$ is the least suspicious value of S, then $\left|\mathrm{AUROC}(S) - \frac{1}{2} - \frac{1}{2}(e_1 - e_0)\right| \leq \frac{1}{2} e_0 e_1$. If every turn's resampling distribution is a point mass at the committed value, $\mathrm{AUROC}(S) = \frac{1}{2}$ however often the commitments are wrong.

Proof. For independent positive and negative draws, both scores equal $s^\star$ with probability $(1 - e_0)(1 - e_1)$; such pairs are ties and contribute exactly $\frac{1}{2}$. All other pairs contribute a value in $[0, 1]$. If $s^\star$ is least suspicious, a pair where only the positive (negative) differs from $s^\star$ contributes 1 (0); only pairs where both differ, with probability $e_0 e_1$, are free. □

On the development corpora (§7.1), the 70B agent's user-dependent commitments (label w_t) have $e_1 = 0.079$ and $e_0 = 0.078$: the corollary caps disagreement's AUROC at 0.576, and its sharp form pins it within $\frac{1}{2} e_0 e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0) \approx \frac{1}{2}$; we measure 0.500, against 0.764 for one noisy reaction. For the less self-consistent 8B agent the cap (0.884) is loose.

Theorem 2 is Blackwell's garbling theorem [3] for a symmetric channel. Let F^ε be feedback generated for the detector: given (u, X, C), which F_t^ε are 0 has a law depending only on (X, C), and each other F_t^ε is $1 - 2w_t$ with its sign flipped independently with probability ε. Let $\eta_\varepsilon = P(L = 1 \mid X, C, F^\varepsilon)$ ($\varepsilon = 0$ is perfect feedback).

Theorem 2 (Less noisy feedback cannot hurt)**.** *Fix the joint law of (u, X, C); in particular, it does not depend on ε. For $0 \leq \varepsilon \leq \varepsilon' \leq \frac{1}{2}$ the ROC curve of η_ε lies pointwise above that of $\eta_{\varepsilon'}$, which lies above that of $\eta_\varnothing$, and $\eta_{1/2} = \eta_\varnothing$ almost surely.*

Proof. If $\varepsilon' = \varepsilon$ there is nothing to show. Otherwise $\varepsilon < \frac{1}{2}$, and flipping each nonzero F_t^ε independently with probability $\delta = (\varepsilon' - \varepsilon)/(1 - 2\varepsilon) \in [0, \frac{1}{2}]$ yields flip rate $\varepsilon + \delta(1 - 2\varepsilon) = \varepsilon'$, hence the joint law of $(u, X, C, F^{\varepsilon'})$. So every detector using $F^{\varepsilon'}$ is a randomized detector using F^ε, and NP gives the first inequality; ignoring F gives the second. At $\varepsilon = \frac{1}{2}$, F^ε is independent of u given (X, C): whether a reaction occurs depends only on (X, C), and every observed sign is a fair coin. □

The premise is essential: if the agent conditions on the same reactions, the law of (X, C) changes with ε and monotonicity can fail. In a two-turn example where the agent copies the reaction to an unconfident answer into a confident one on a correlated attribute without feedback, $\mathrm{AUROC}(\eta_\varepsilon)$ is 0.50 at $\varepsilon = 0$ and 0.71 at $\varepsilon = 0.25$. In PGE, Theorem 2 thus covers regenerating feedback for the detector with the agent's trajectories fixed.

The label-event posterior. Let $\kappa_t = (a_t, v_t, b(p_t))$ be the detector-visible *cell*, where $b(\cdot)$ buckets confidence, and $\mathscr{H}_t = (X_t, c_{1:t}, F_{<t}, w_{<t})$ the history up to the t-th commitment with the earlier errors. (A1) Given $\mathscr{H}_t$, $w_t \sim \mathrm{Bernoulli}(\pi_{\kappa_t})$. (A2) Given $(\mathscr{H}_t, w_t)$, $F_t = 0$ with a probability that depends only on a_t, and otherwise $F_t = 1 - 2w_t$ with its sign flipped with probability ε. Up to a factor free of w_t, the likelihood of F_t is $\ell_{w_t}(F_t)$ with $\ell_w(0) = 1$, $\ell_1(-1) = \ell_0(+1) = 1 - \varepsilon$ and $\ell_1(+1) = \ell_0(-1) = \varepsilon$. (A1) excludes error cascades and intake information beyond the cell, but lets the agent choose cells after seeing feedback.

Proposition 1 (Exact posterior)**.** *Under (A1)–(A2), $P(w_{1:T} \mid X, C, F) = \prod_t q_t^{w_t}(1 - q_t)^{1 - w_t}$ with $q_t = \pi_{\kappa_t}\ell_1(F_t) / \big(\pi_{\kappa_t}\ell_1(F_t) + (1 - \pi_{\kappa_t})\ell_0(F_t)\big)$. Hence $\eta_\varepsilon = P(L = 1 \mid X, C, F)$ is 0 if $H = \varnothing$ and otherwise the Poisson-binomial tail $P\big(\sum_{t \in H} B_t \geq \lceil \rho |H| \rceil\big)$, $B_t \sim \mathrm{Bernoulli}(q_t)$ independent, computable in $O(|H|^2)$. It is Bayes-optimal under (A1)–(A2) with the true priors and channel; the implementation plugs in estimates.*

Proof. Write the joint law in the order $c_1, w_1, F_1, c_2, \ldots$. The policy and the questions see u only through X, so $p(X, C, F, w) = A(X, C, F) \prod_t \pi_{\kappa_t}^{w_t}(1 - \pi_{\kappa_t})^{1 - w_t} \ell_{w_t}(F_t)$, where A collects the intake prior, the questions, the policy and the w-free factors of (A2). Normalizing over $w \in \{0, 1\}^T$ gives independent $\mathrm{Bernoulli}(q_t)$. H is fixed by C, L depends on w only through $\sum_{t \in H} w_t$, and NP gives optimality. □

The same proof gives the posterior of L_A (the tail over H_A), and holds for any channel $P(F_t \mid \mathscr{H}_t, w_t) = \ell_{w_t}(F_t)$ that may depend on a_t, as in the general-channel detector. Under (A1) this user's reactions enter only through $\ell_1(F_t)/\ell_0(F_t)$;

setting these to 1 leaves the population base rates π_κ (the prior-only ablation). Cascades violate (A1), but a simple cascade model does not help (§7.6).

Proposition 2 (Priors are identified where feedback exists)**.** *Assume (A2) and let $\pi_\kappa = P(w_t = 1 \mid \kappa_t = \kappa)$. Let n_κ count the commitments in cell κ that received a reaction ($F_t \neq 0$) and $\hat{r}_\kappa$ the fraction contradicted, and assume their errors are independent given n_κ (implied by (A1), or by i.i.d. sessions in which each attribute is committed at most once). Then the contradiction rate is $r_\kappa = \varepsilon + (1-2\varepsilon)\pi_\kappa$, and for known $\varepsilon < \frac{1}{2}$, $\hat{\pi}_\kappa = (\hat{r}_\kappa - \varepsilon)/(1-2\varepsilon)$ needs no labels and, given n_κ, with probability at least $1-\delta$ [9],*

$$|\hat{\pi}_\kappa - \pi_\kappa| \;\le\; \frac{1}{1-2\varepsilon}\sqrt{\frac{\log(2/\delta)}{2n_\kappa}};$$

projecting onto $[0,1]$ does not increase the error. Cells on attributes that never receive feedback are not identified. If ε is unknown, the reactions depend on (π,ε) only through $(r_\kappa)_\kappa$, so every $\varepsilon' \in [0,\frac{1}{2})$ with $\varepsilon' \le \min_\kappa \min(r_\kappa, 1-r_\kappa)$, with $\pi'_\kappa = (r_\kappa - \varepsilon')/(1-2\varepsilon')$, fits equally well.

One scalar anchor (ε, or one cell's $\pi_\kappa \neq \frac{1}{2}$) resolves this; we choose ε by training-split label likelihood. Unidentified cells get the bucket prior pooled over the other attributes. The bound covers the raw estimate, not the implementation's clipping to $[0.01, 0.99]$ and shrinkage toward coarser cells (value $\to$ attribute $\to$ bucket).

5 Detectors

PGE posterior. The score is a plug-in estimate of the posterior of Proposition 1. Each confident commitment is assigned to a cell $\kappa = (\text{attribute}, \text{value}, \text{confidence bucket})$ with buckets $[0.6, 0.65)$, $[0.65, 0.75)$, $[0.75, 1]$. Cell priors are estimated from training reactions by the method of moments (Proposition 2), shrunk toward attribute-level and bucket-level estimates (strength $k = 5$); cells on the two attributes that never receive reactions are not identified and get the pooled bucket-level prior. Our pre-registered primary detector, V6, uses a general channel: the contradiction rates $r_w = P(F = -1 \mid w)$ are chosen on training labels over a grid ($r_1 > r_0 + 0.1$), priors are $\pi = (\text{rate} - r_0)/(r_1 - r_0)$, and the confirmation rates $g_w = P(F = +1 \mid w)$ are fit by least squares; the symmetric flip channel (V3, with flip rate ε chosen on training labels) is the special case $r_0 = g_1 = \varepsilon$, $r_1 = g_0 = 1 - \varepsilon$. Proposition 1 covers this channel, since its proof uses only $P(F_t \mid w_t)$; the channel scale is chosen on training labels because reactions alone do not identify it (Proposition 2). For L_U the label event is restricted to commitments on attributes that depend on a hidden field anywhere in the schema, a set the detector can know without the clinic's rules. V6 is therefore the posterior of a proxy event (L_A of Section 4 with A the schema-level set), not of L_U: on the development corpora the two events differ in most sessions and the labels disagree on a fifth to a quarter of them. *Prior-only* (V6−prior) keeps the fitted priors and ignores the test user's reactions; V6 minus V6−prior is the feedback increment.

Baselines. SelfCheckGPT and semantic entropy on $k = 3$ resamples per turn at temperature 0.9 (exact-match clustering, adequate for categorical answers; for L_U computed on the user-dependent turns); Chain-of-Verification with the agent as its own verifier [7]; mean stated confidence (sign chosen on the training split); two feedback statistics (fraction of contradicted confident commitments, fraction of contradicted turns); a logistic regression on 14 reasoning-graph features; a two-layer GCN [13]; five block-model graphon likelihoods [19], of which we report the best on test, favouring the baseline. Two supervised baselines see exactly the posterior's inputs: *ctx-LR* is an L_2 logistic regression ($C = 0.1$, fixed) on counts of confident commitments per (attribute, value) cell, $|H|$, the number of turns and mean confidence, with no reactions and no samples; *ctx-LR+F* adds counts per (attribute, value, reaction) cell.

6 Experimental Design

Environment. A simulated imaging clinic. Five coarse intake answers are shown to the agent; four hidden fields (days since treatment, allergy severity, kidney function, anxiety) are not. Each session asks for 6–12 of 12 attributes in dependency order. Six attributes can depend on a hidden field, but only for some intakes (a seventh, `transport_assistance`, depends on one in the schema but is always fixed by the intake). Ground truth is a deterministic function of intake and hidden fields; the hidden fields are independent draws from a fixed prior. The agent is a real LLM; only the patient is simulated.

Reactions. In the main corpora the reaction to a commitment is its correctness bit flipped with probability $\varepsilon = 0.25$, absent for two attributes the patient never reacts to. In the *LLM-patient* corpus a second model (Llama-3.1-8B) role-plays the patient: it is given its own history but not the clinic's decision, one of three personas (attentive, trusting, anxious), and the agent's commitment as a hedged plain-language sentence, and replies in one or two sentences. A third model that sees only the sentence and the reply labels it as agreement, contradiction or neither. The agent reads the patient's verbatim replies. An *informed* variant also tells the patient the clinic's decision.

Development and confirmation. All method development used two development corpora (Llama-3.3-70B, 600 sessions; Llama-3.1-8B, 200). An independent audit of these results found, among other issues, that the agent had been shown a corrupted feedback display (no reaction appeared as acceptance, and below $p = 0.5$ acceptance and contradiction appeared swapped), that resamples covered only five attributes, and that a feedback-free classifier nearly matched the posterior on L. Before generating new data, we committed an analysis plan to the

Table 1: Per-commitment check of Corollary 1 on user-dependent commitments. e_1, e_0: probability that the three resamples do not all agree, given that the commitment is wrong / right. AUROC for predicting *that* commitment's error.

Agent	wrong	all agree	e_1	e_0	Disagreement	One reaction
Llama-70B	??%	??%	??	??	??	??
Llama-8B	??%	??%	??	??	??	??
Qwen-72B	??%	??%	??	??	??	??
GPT-4o-mini	??%	??%	??	??	??	??
Mistral-24B	??%	??%	??	??	??	??

code repository [24]: labels, detectors (V6 primary), every comparison, the statistics, and the wording each outcome licenses. Two amendments, also committed before any confirmatory data existed, added the analyses of Sections 7.3 and the supervised feedback increment. The confirmatory corpora use a fresh random seed (no session seen in development), show the agent the raw reaction, draw three resamples on every turn from that turn's own context, and cover five agents: Llama-3.3-70B, Llama-3.1-8B, Qwen2.5-72B, GPT-4o-mini and Mistral-Small-3.1-24B, 600 sessions each. Every result below is confirmatory unless marked as development.

Statistics. Splits are 70/30 by session for three split seeds; detectors are fit on the training split and score the test split once; AUROC is averaged over seeds. Differences use a paired bootstrap (2,000 resamples) clustered by session, since test sets of different seeds overlap. Pooled estimates average per-agent differences, with sessions resampled within agent. A difference is called significant when its 95% interval excludes zero; we do not correct for multiplicity and report every registered comparison.

Oracle-information analyses. Because the simulator's rules and hidden-field prior are known, we can compute three exact quantities. The *population risk* $\bar{\eta} = \sum_h P(h)\, L_U(C,h)$ is the probability that the session's commitments would be wrong for a random user with the same intake. $\eta_\varnothing = P(L_U = 1 \mid X, C)$ adds the reactions the agent was shown (all but the last); by Theorem 1 it bounds every detector computed from the agent's context and its resamples. η_F adds the last reaction and bounds every detector of commitments and reactions. The agent's policy does not involve the hidden facts, so its factors cancel and no agent model is needed. These are not deployable detectors; they locate the information. We also give V6 and ctx-LR(+F) the same oracle knowledge of which commitments are user-dependent in each session (V6-session, LR-session).

7 Results

7.1 Sampling cannot see the user

Table 1 shows the mechanism. On commitments whose truth depends on undisclosed facts, agents that are self-consistent repeat the same answer whether it is right or wrong (e_0 and e_1 small and close, so by Corollary 1 the AUROC of disagreement lies within $\frac{1}{2}e_0e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0)$), so disagreement carries little information, while one reaction with 25% noise carries a great deal. The development corpora show the same pattern (70B: all three resamples agree on 92.1% of these commitments, disagreement AUROC 0.500, one reaction 0.764). At the session level (Table 2), SelfCheckGPT and semantic entropy score ?? and ?? pooled over agents ([PENDING: H1 licensed claim]). Ten resamples instead of three give ?? against ?? for the 70B agent. The contrast with tasks whose errors are computational is sharp: on AgentBench database questions [17], where errors vary across samples, SelfCheckGPT reaches 0.712 with the same 70B agent (development).

7.2 Detecting user-specific hallucination

Table 2 gives the confirmatory results and Table 3 the registered hypotheses, pooled over agents. [PENDING: H2/H2b/H3/H4/H4b paragraph, worded by the registered rules]

7.3 Where the information is

Table 4 locates the signal. The population risk $\bar{\eta}$ alone is far above every sampling detector: much of L_U is visible in *which* values an agent commits to, given the intake. The user's reactions add $\eta_F - \bar{\eta}$ beyond it (H5). Almost all of that information is already in the conversation the agent sees ($\eta_\varnothing$ is close to η_F), so the failure of sampling detectors is not that the information is absent from their input; it is that agreement between resamples does not read it. [PENDING: H6 sentence: posterior vs supervised given the same information] Without the oracle knowledge, V6 targets a schema-level set of attributes that is only a proxy for the label's support, which costs it [PENDING: gap] against V6-session: knowing *which* commitments concern undisclosed facts matters as much as how the evidence is combined.

7.4 Natural-language reactions

[PENDING: LLM-patient paragraph: audit (P(-1|wrong), P(-1|right), lift for profile and informed arms), detection V6 / V6-prior / ctx-LR / ctx-LR+F on both labels]

7.5 Choosing what to ask

[PENDING: measured clarification: posterior@1 vs no-question, random, lowest-confidence; posterior@2 vs detailed questions and the full-rerun control; questions per

Table 2: Session AUROC on the user-specific label L_U (mean of three splits; confirmatory corpora, 600 sessions per agent). The composite label L is in the repository. CoVe was run for the 70B agent only.

	Detector	Llama-70B	Llama-8B	Qwen-72B	GPT-4o-mini	Mistral-24B
Model only	SelfCheckGPT	??	??	??	??	??
	Semantic entropy	??	??	??	??	??
	CoVe	??	–	–	–	–
Commitments	Mean confidence	??	??	??	??	??
	ctx-LR	??	??	??	??	??
	V6−prior	??	??	??	??	??
Commitments + reactions	Contradicted confident	??	??	??	??	??
	Graph-feature LR	??	??	??	??	??
	GCN	??	??	??	??	??
	Best graphon	??	??	??	??	??
	ctx-LR+F	??	??	??	??	??
	PGE posterior (V6)	??	??	??	??	??
	L_U positive rate	??%	??%	??%	??%	??%

Table 3: Registered hypotheses on L_U, pooled over five agents (stratified session bootstrap; *: 95% CI excludes 0).

	Comparison	Estimate [95% CI]
H2	V6 − V6−prior (feedback increment)	??
H2b	ctx-LR+F − ctx-LR (feedback increment)	??
H3	V6 − ctx-LR	??
H4	V6 − SelfCheckGPT	??
H4	V6 − semantic entropy	??
H4b	ctx-LR+F − SelfCheckGPT	??
H4b	ctx-LR+F − semantic entropy	??
–	V6 − ctx-LR+F	??

Table 4: Oracle-information analysis on L_U (AUROC). $\bar{\eta}$: population risk of the commitments; $\eta_\varnothing$: Bayes given the conversation the agent saw (Theorem 1's bound for context-and-sample detectors); η_F: Bayes given all reactions. V6-session and LR-session+F are told which commitments are user-dependent in each session.

Agent	$\bar{\eta}$	$\eta_\varnothing$	η_F	V6-session	LR-session+F
Llama-70B	??	??	??	??	??
Llama-8B	??	??	??	??	??
Qwen-72B	??	??	??	??	??
GPT-4o-mini	??	??	??	??	??
Mistral-24B	??	??	??	??	??

Pooled: $\eta_F - \bar{\eta} =$?? (H5); V6-session − LR-session+F = ?? (H6)

session; primary outcome wrong confident user-dependent commitments]

7.6 Development-phase robustness

On the development corpora we stress-tested the posterior by replacing the reactions seen by the detectors (not the agent) with channels it does not assume: symmetric flips at 0.10–0.40, asymmetric flips, attribute-dependent noise, extra missingness, and a "lazy" user who contradicts 60% of wrong commitments and never confirms correct ones (labels L, ten channels). The symmetric posterior V3 collapses under the lazy user (0.425 for 70B, 0.576 for 8B): it reads the absence of confirmations as evidence of correctness. The general-channel V6 recovers the channel from data and repairs this (0.857 and 0.855). Across all ten channels V6 has the best point estimate on both agents; with session-clustered intervals its lead over the best baseline is significant in six channels for 70B and two for 8B, so the robustness claim is about consistency, not significance. Structure-based detection does not help: graphon likelihoods score between 0.43 and 0.58 on every development corpus and label, even with the best variant chosen on test, and exact inference in a simple cascade model over the dependency graph (errors depend on attribute, bucket, and whether any parent is wrong) is no better than independence (0.732 vs. 0.734 without value conditioning).

8 Limitations

Simulation. The agents are real, but the patient is simulated: by a noisy correctness bit in the main corpora and by an LLM in one corpus. The oracle analyses rely on the simulator's known rules and prior, which a deployment does not have. *One domain.* All private-attribute results come from one clinic schema; a second domain with hidden user preferences (e.g. a retail or travel assistant) is the most important next test. *Labels.* Labels are programmatic and noise-free by construction; there is no human evaluation. *Supervision.* The posterior and the supervised baselines are fit on labelled training sessions; the posterior can run without labels when the channel is known, which we only test in development. *White-box detectors.* We did not run hidden-state probes; Theorem 1 predicts they share the within-context limit, but they could exploit population risk as ctx-LR does. *Theory.* Proposition 1 is exact only un-

der (A1)–(A2) with the true priors and channel, and the detector is a plug-in estimate; Theorem 2 covers reactions regenerated for the detector with the agent's trajectories fixed, not the protocol's own noise, because the agent conditions on the reactions it is shown.

9 Conclusion

When an agent invents a fact about its user, its own samples cannot tell which of the users sharing its context it is wrong about. Some of the risk is visible in what the agent commits to; the rest is visible only through the user's reactions. Making the agent state its assumptions as probabilistic commitments and recording the user's reactions makes both available, and a Bayes analysis in a simulator with known truth shows how much each is worth. [PENDING: one-sentence summary of the confirmatory outcome for the posterior and for clarification]

Reproducibility. Code, all trajectories, the preregistered plan with its amendments and commit history, every analysis script, and a ledger of all runs including discarded variants are at [PENDING: anonymous repository link].

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